TOTHBOT

KRAKEN SYSTEM ARCHITECTURE OVERVIEW

The Complete System — All Architectural Decisions + All Flow Diagrams

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PURPOSE: This document captures all architectural decisions, FP/DP analyses, and system specifications from the TothBot Kraken architecture sessions of March 2026. Includes all six system flow diagrams showing connections between Kraken, TothBot, and CIATS as one unified organism.

Diagrams in this document (dv1_6 — all updated):

Diagram 1 — Kraken API Connection Architecture (dv1_6: amend_order primary, batch_cancel, WS ping A-7, zombie A-8, balances seq A-9, per-pair status, L3 URL corrected)

Diagram 2 — TothBot Trading Pipeline (dv1_6: per-pair instrument status check added before Gate 1)

Diagram 3 — Exit Architecture: Ground Zero Redesign (dv1_6: title updated)

Diagram 4 — Regime Taxonomy: 6-Regime Grid + Two-Level Tagging (dv1_6: title updated)

Diagram 5 — CIATS Architecture: Data Flows and Improvement Cycle (dv1_6: title updated)

Diagram 6 — Complete System: One CIATS-Owned Organism (dv1_6: all changes integrated)

1. System Identity and Mission

1.1 What TothBot Is

TothBot is an automated cryptocurrency spot trading system on Kraken exchange. It runs 24/7/365 on a Hetzner VPS, processes 5-minute OHLC candle close events as its system clock, and executes long-only trades using a momentum-based signal engine.

TothBot is not an external system connected to Kraken — it IS Kraken's event processor. Every action is a response to a Kraken push event. No timers. No polling. Idle and silent until Kraken speaks.

1.2 The One Organism Principle

Three functional zones, one organism:

• Kraken: generates events, executes orders, holds resting TP limit orders and off-book emergency SL stop orders on matching engine

• TothBot: processes Kraken events, executes trading strategy, manages positions

• CIATS: learns from event records, continuously improves processing rules

PRIMARY MISSION: Maximize profit. Minimize loss. With an emphasis on minimizing loss. Loss prevention takes priority over profit generation. Everything else is secondary.

1.3 Operating Environment

Parameter	Value
Exchange	Kraken Spot (Long Module — Short deferred)
Language	Python 3.12.3

VPS	Hetzner CPX22 — 87.99.141.44 — Ashburn VA (Kraken Spot: AWS us-west-2 Oregon) ★ Evaluate Hillsboro OR [dv1_6 IMP-01]
System Clock	Kraken OHLC 5-minute candle close event (WS v2 push)
Fee Baseline	Maker 0.16% / Taker 0.26% (CIATS monitors actuals)
Operation	24/7/365 — CIATS may add stand-down periods via PDCA if data supports
Starting Portfolio	$1,000 Long Module
Sacred constraint	Net 1:1.5 R:R — HARDCODED — the only number not owned by CIATS

2. Kraken API Connection Architecture

All connection points verified across 3 independent analysis passes (15 sources) plus 2 deep validation sessions (144+ sources total) plus 10-search FP/DP Kraken validation [dv1_6] (126 sources). The diagram below shows the complete interface between TothBot and Kraken.

2.1 Key Connection Points

• Public WS v2: instrument (asset specs + per-pair status cache ★ dv1_6) · status (engine state) · ohlc(5m) (system clock) · ohlc(60) (1H HTF cache for Gate 4 — dv1_5) · ticker (price data, event_trigger configurable)

• Private WS v2 inbound: executions (ratecounter:true — A-1, snap_orders:true on reconnect — A-3) · balances (real-time settlement, sequence gap detection ★ A-9 dv1_6)

• Private WS v2 outbound: add_order (entry) · amend_order (primary amendment — edit_order DEPRECATED ★ dv1_6) · batch_add (SL+TP atomic) · cancel_order (exit leg cleanup) · batch_cancel (Full Halt ★ dv1_6) — all messages carry req_id (A-2), cl_ord_id (A-2), stp_type:cancel-newest (A-4)

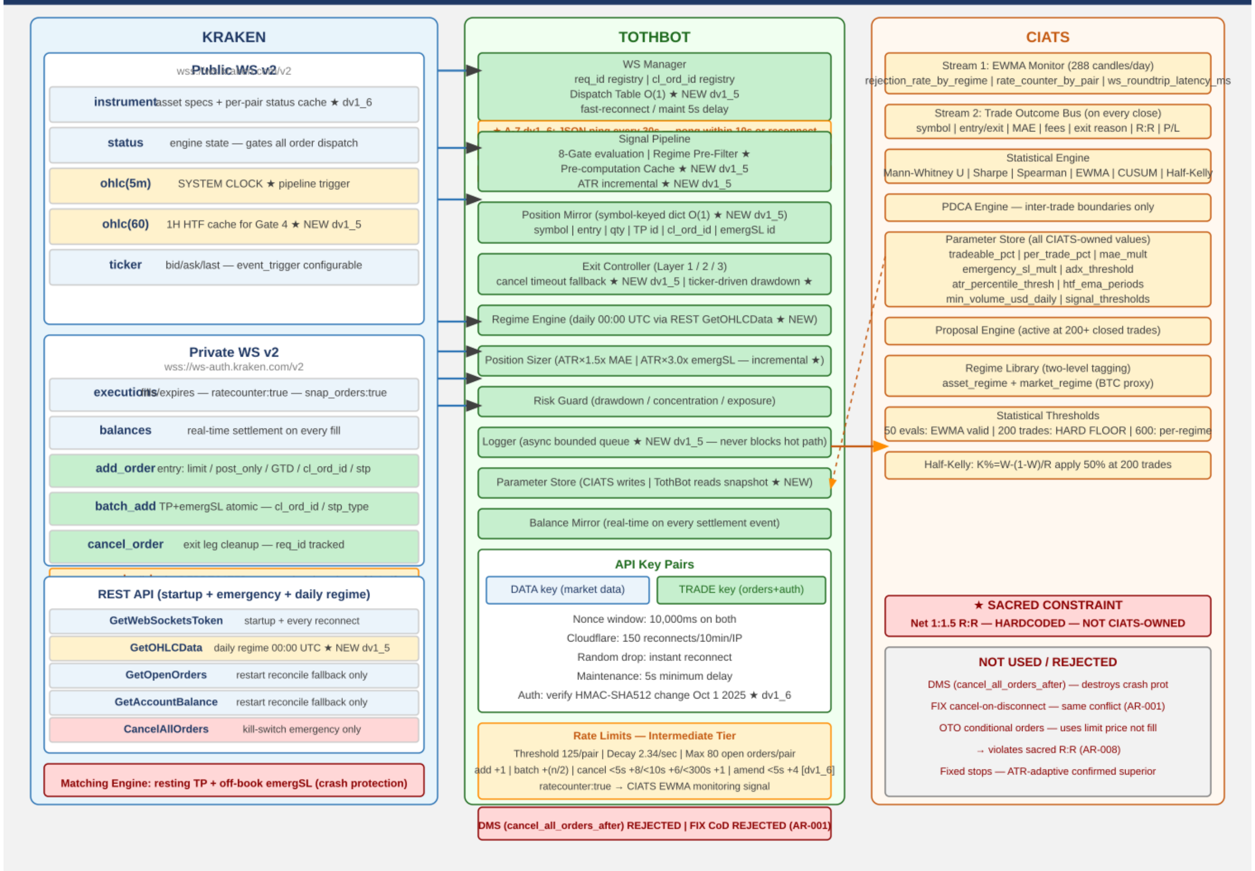
• REST: GetWebSocketsToken (startup/reconnect) · GetOHLCData (daily regime 00:00 UTC — dv1_5) · GetOpenOrders + GetAccountBalance (restart fallback) · CancelAllOrders (kill switch)

• Matching Engine: resting TP limit orders and off-book emergency SL stop orders survive any TothBot failure — crash protection

• cancel_all_orders_after (WS v2 DMS): EVALUATED AND REJECTED — destroys crash protection (AR-001)

• FIX API cancel-on-disconnect: EVALUATED AND REJECTED — same architectural conflict (AR-001)

DIAGRAM 1 — Kraken API Connection Architecture (dv1_6)



2.2 Four Kraken Engine States

State	TothBot Behavior
online	All orders. Signal pipeline runs normally.
post_only	TothBot limit+post_only entries ARE valid. Trading continues.
cancel_only	No new entries. Exit Controller can cancel. Resting TP and off-book SL stop orders active.
maintenance	No orders, no cancels, no matching. Resting TP and off-book SL stop orders active on engine.

2.3 API Rate Limits — Intermediate Tier (C-2)

Rate limits are shared across REST, WebSockets, and FIX — there is no separate per-protocol budget.

Parameter	Intermediate Tier Value
Rate counter threshold (per pair)	125
Rate counter decay (per second)	2.34
Max open orders (per pair)	80
Add order fixed cost	+1
Amend order cost ★ dv1_6 (PRIMARY)	+1 fixed, +3 (<5s), +2 (<10s), +1 (<15s) — 43% cheaper than edit_order for <5s orders
Edit order cost (DEPRECATED ★ dv1_6)	+1 fixed, +6 (<5s), +5 (<10s), +4 (<15s) — DO NOT USE, use amend_order
Batch add fixed cost	+(n/2) where n = orders in batch
Cancel order cost (< 5s old)	+8
Cancel order cost (< 10s old)	+6

Cancel order cost (< 300s old)	+1
Batch cancel cost ★ dv1_6	Cumulative sum of individual order cancel costs. If cumulative exceeds threshold, requests still accepted.

Operational note: At Layer 2 MAE exit, the Exit Controller cancels both TP and emergSL before market-selling. Cancel cost depends on how long those orders have been resting. The 125 threshold per pair with 2.34/sec decay provides substantial headroom at TothBot's 5-minute candle frequency. Full rate limit specification and CIATS monitoring hooks documented in 0411003.

2.4 WS Operational Requirements (A-1 through A-9)

A-1 — Rate counter monitoring via executions channel:

The executions subscription accepts ratecounter:true. Real-time rate counter value per pair included at zero additional connection cost. CIATS Stream 1 EWMA Monitor adds rate_counter_by_pair as third monitoring signal. Assign to 0511003 and 0211001.

A-2 — req_id and cl_ord_id on all outbound WS messages:

All WS v2 outbound messages carry req_id (client-specified integer). All add_order and batch_add messages additionally carry cl_ord_id (up to 18 characters). TothBot assigns cl_ord_id at dispatch time before Kraken response. Position Mirror records cl_ord_id immediately on dispatch — no async gap. Cancel operations may reference cl_ord_id. WS Manager maintains a pending-request registry and alerts on unacknowledged messages after timeout. Assign to 0511001, 0511003, 0511005.

A-3 — Round-trip latency monitoring and WS reconciliation:

All WS v2 responses include time_in and time_out to nanosecond precision. Logger records these fields. CIATS EWMA Monitor adds ws_roundtrip_latency_ms. executions channel supports snap_orders:true on subscription — preferred over REST GetOpenOrders for restart reconciliation. REST remains as cold-start fallback. Assign to 0511001, 0511006, 0211001.

A-4 — stp_type:cancel-newest on all add_order calls:

All add_order and batch_add calls include stp_type:cancel-newest. Self Trade Prevention prevents TothBot from inadvertently filling against its own resting orders. Zero-cost prevention. Assign to 0511003.

A-5 — Reconnect behavior distinction:

Two distinct reconnect behaviors: (1) Random drop — reconnect instantly, up to a handful of attempts; (2) After confirmed trading engine maintenance — minimum 5-second delay before reconnect. Both behaviors confirmed per Kraken documentation. Assign to 0511001.

A-6 (dv1_5) — OTO conditional orders — REJECTED:

OTO uses limit price not actual fill price — violates sacred R:R. Creates Position Mirror race condition. batch_add on fill confirmed correct. See AR-008.

A-7 (dv1_6) ★ — WS Ping / Keepalive Strategy:

Kraken WS server disconnects after ~60 seconds of inactivity. Cloudflare idle timeout ~100 seconds. WS Manager MUST send {"method":"ping"} on each active connection every 30 seconds. Expects {"method":"pong"} within 10 seconds. No pong within 10 seconds = dead connection. Reconnect immediately. This is application-level ping, distinct from TCP keepalive. Assign to 0511001, 1011002.

A-8 (dv1_6) ★ — Zombie Connection Detection:

A WS connection can appear alive (ping/pong passing) while delivering no real market data. WS Manager MUST track last_real_data_time per connection using time.monotonic(). Only actual market data events reset this timestamp — heartbeat messages alone do NOT reset it. If elapsed > 90 seconds: log ZOMBIE_CONNECTION_DETECTED, alert operator, trigger reconnection. Assign to 0511001, 1011002.

A-9 (dv1_6) ★ — Balances Channel Sequence Gap Detection:

Every balances channel message carries a sequence integer. WS Manager MUST track last-seen sequence. Any gap (skip > 1) indicates a missed ledger event. On gap: alert immediately, trigger REST GetAccountBalance reconciliation. Missed balance events cause incorrect available capital calculation — direct loss risk. Assign to 0511001, 1011002.

2.5 Data Architecture: Pre-Computation Cache (NEW dv1_5 — I-2)

HOT PATH RULE: The 5-minute pipeline hot path reads cached values ONLY. No computation occurs during pipeline evaluation. All indicators maintained as incremental running calculations updated on each event trigger.

Value	Pre-computed by	Update Trigger	Cache Read On
Regime classification (all pairs)	Regime Engine	Daily 00:00 UTC — REST GetOHLCData [CRITICAL: exclude response[-1] dv1_6]	Every 5m pipeline eval (Gate 3 + Gate 6)
1H EMA(20) and EMA(50) (all pairs)	Signal Pipeline	1H candle close — ohlc(60) WS event (NEW dv1_5)	Gate 4 (1H HTF Confirmation)
ATR(14) per pair	Signal Pipeline	Each 5m candle close (incremental — I-9)	Gate 8 Position Sizer + Exit Controller
24h liquidity per pair (USD)	Signal Pipeline	Daily ticker or daily OHLC pull	Gate 2 (Liquidity Gate)
Per-pair instrument status dv1_6	WS Manager	instrument channel events (live)	Gates 1/2 — skip non-online/post_only pairs

3. TothBot Trading Pipeline

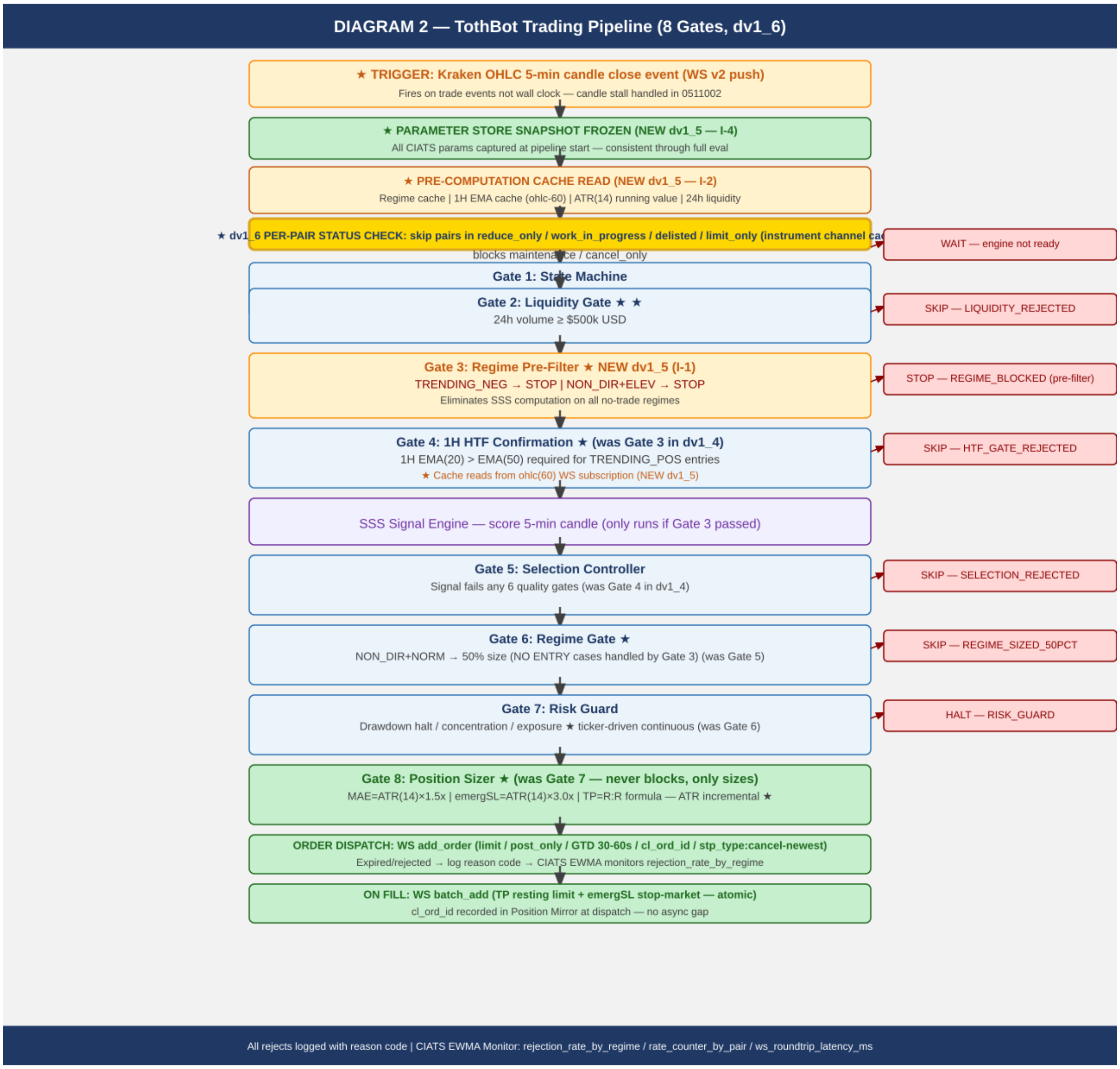
The pipeline fires on every 5-minute OHLC candle close event. It is synchronous and event-driven. All 8 gates incorporate FP/DP validated improvements. Gate 3 (Regime Pre-Filter) is new in dv1_5. Per-pair instrument status check new in dv1_6.

NOTE — OHLC fires on trade events, not wall clock: The OHLC channel publishes an updated candle only when a trade occurs in that interval. If no trades occur after an interval border, no message is sent until the next trade. The Liquidity Gate ($500k/day minimum) materially reduces this risk. The same stall behavior applies to ohlc(60). TothBot must handle the possibility of a delayed or stalled system clock. To be addressed in 0511002 Signal Pipeline Specification.

3.1 Gate Summary

Step	Name	Blocks / Acts When
Snapshot	Parameter Store Frozen (NEW dv1_5 — I-4)	CIATS parameter values captured as read-only snapshot at pipeline start. Consistent throughout full eval. CIATS writes applied only at inter-trade boundaries.
Cache	Pre-computation Cache Read (NEW dv1_5 — I-2)	Hot path reads: regime cache | 1H EMA cache (from ohlc-60) | ATR(14) running value | 24h liquidity | per-pair status. Zero computation on hot path.
★ dv1_6	Per-pair Instrument Status Check	Skip pairs in reduce_only / work_in_progress / delisted / limit_only state. Only online or post_only pairs proceed into pipeline. Prevents wasted rate counter on rejected orders. Source: instrument channel cache.
Gate 1	State Machine	Kraken engine is maintenance or cancel_only — WAIT
Gate 2 ★	Liquidity Gate	Asset 24h volume < $500k USD — SKIP: LIQUIDITY_REJECTED
Gate 3 ★ NEW	Regime Pre-Filter (NEW dv1_5 — I-1)	TRENDING_NEG (any vol) or NON_DIR+ELEVATED — STOP: REGIME_BLOCKED. Eliminates SSS computation on all no-trade regimes. Definitive NO ENTRY check moved before expensive signal computation.
Gate 4 ★	1H HTF Confirmation (was Gate 3)	1H EMA(20) <= EMA(50) for TRENDING_POS entries — SKIP: HTF_GATE_REJECTED. Cache read from ohlc(60) WS subscription (NEW dv1_5).
SSS	Signal Engine (SSS)	Scores 5-min candle. Only reached if Gate 3 pre-filter passed — wasted computation eliminated on no-trade regimes.
Gate 5	Selection Controller (was Gate 4)	Signal fails any of 6 quality gates — SKIP: SELECTION_REJECTED

Gate 6 ★	Regime Gate (was Gate 5)	NON_DIR+NORMAL: 50% position size. All NO ENTRY cases already blocked by Gate 3. Gate 6 now only handles position sizing for the one tradeable non-trending regime.
Gate 7	Risk Guard (was Gate 6)	Drawdown halt triggered, concentration exceeded, exposure limit. ALSO: ticker-driven continuous drawdown monitoring for held positions between candle closes (NEW dv1_5 — I-7).
Gate 8 ★	Position Sizer (was Gate 7)	ATR(14) x 1.5x MAE, ATR(14) x 3.0x emergSL. Never blocks — only sizes. ATR maintained as incremental running calculation (NEW dv1_5 — I-9).

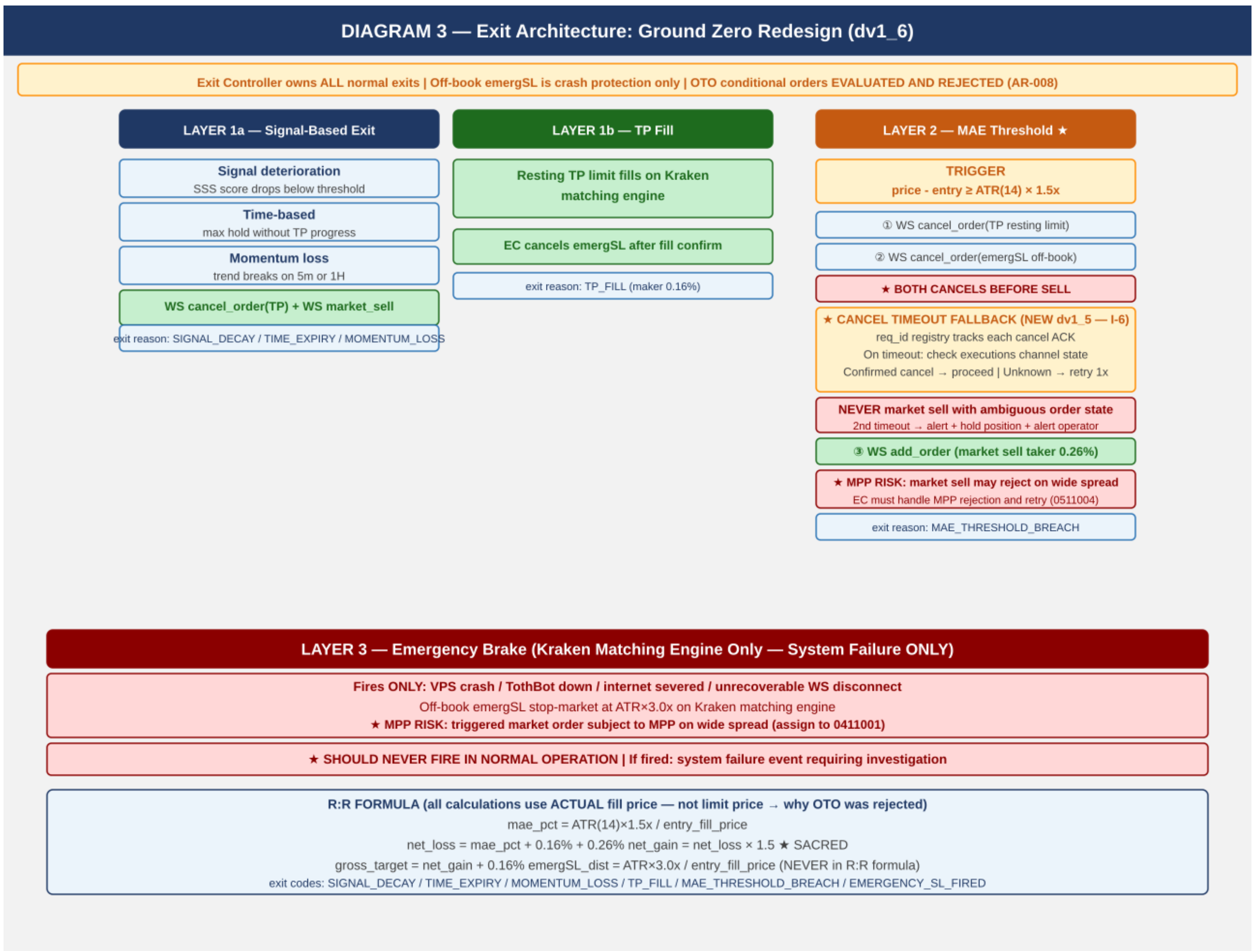


4. Exit Architecture — Ground Zero Redesign for Kraken

This exit architecture was rebuilt from zero for Kraken. It eliminates all competing exit strategies. The Exit Controller owns all normal exits. The Kraken off-book emergency SL stop order is crash protection only. OTO conditional orders were evaluated and rejected — batch_add on fill is confirmed correct (see AR-008).

4.1 Three-Layer Summary

Layer	Name	When It Fires / Behavior
1a	Signal-Based Exit	Signal decay / time expiry / momentum loss. WS cancel TP + market sell.
1b	TP Fill	Resting TP limit on Kraken matching engine fills. EC cancels emergSL after fill confirm.
2 ★	MAE Threshold	price - entry >= ATR(14) x 1.5x. SEQUENCE CRITICAL: (1) cancel TP (2) cancel emergSL THEN (3) market sell. ★ CANCEL TIMEOUT FALLBACK (NEW dv1_5 — I-6): req_id registry tracks each cancel ACK. On timeout: check executions channel state. Confirmed cancelled: proceed. State unknown: retry once. Second timeout: ALERT + hold position. NEVER issue market sell with ambiguous order state. ★ MPP RISK (C-1 dv1_4): market sell may reject on wide spread. Assign to 0511004.
3 ★	Emergency Brake	VPS/TothBot/internet failure ONLY. Off-book emergSL stop order at ATR(14) x 3.0x on Kraken matching engine. NEVER in normal operation. ★ MPP RISK (C-1 dv1_4): triggered market order subject to MPP on wide spread. Assign to 0411001.



4.2 R:R Formula

mae_pct = ATR(14) x 1.5x / entry_fill_price (CIATS-owned 1.5x multiplier, uses ACTUAL fill price)

net_loss = mae_pct + 0.16% + 0.26% (entry maker + MAE taker exit)

net_gain = net_loss x 1.5 (SACRED — hardcoded)

gross_target = net_gain + 0.16%   (TP maker exit)

emergSL_dist = ATR(14) x 3.0x / entry_fill_price   (NEVER in R:R formula — outside envelope)

NOTE: All calculations use actual fill price, not limit price. OTO conditional uses limit price — this is why OTO was rejected (AR-008).

5. Regime Taxonomy

The six-regime grid is the foundation of CIATS regime-segmented analysis. Every trade is tagged with both asset_regime and market_regime (BTC proxy). Computed daily at 00:00 UTC.

5.1 Computation

DATA SOURCE (NEW dv1_5 — FLAG 2 — FP/DP VERIFIED): Daily candles via REST GetOHLCData called once per monitored pair at 00:00 UTC. CRITICAL dv1_6: The last entry (response[-1]) is ALWAYS the current not-yet-committed candle — MUST be excluded from ALL calculations (ADX, EMA, ATR). Use only response[:-1]. Including the uncommitted candle corrupts all regime indicators. See GAP-03 / AR-017 / 1011012 Regime_Engine_Coding_Spec.

Directional state: ADX(14) + EMA(20) vs EMA(50) on DAILY candles (from REST GetOHLCData — committed candles response[:-1] ONLY)

Volatility state: ATR(14) percentile rank in 50-day rolling window

TRENDING_POSITIVE: ADX(14) > 25 AND EMA(20) > EMA(50)

TRENDING_NEGATIVE: ADX(14) > 25 AND EMA(20) < EMA(50)

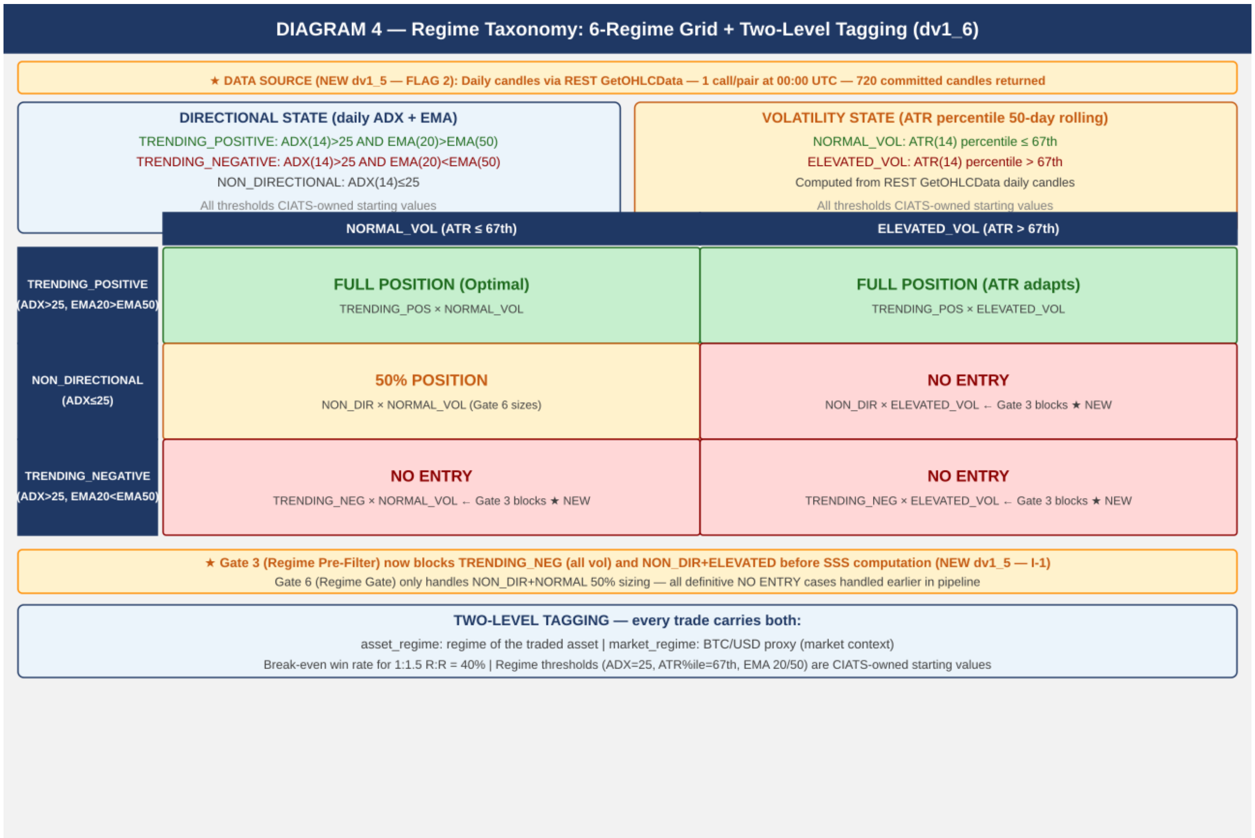
NON_DIRECTIONAL: ADX(14) <= 25

NORMAL_VOL: ATR percentile <= 67th

ELEVATED_VOL: ATR percentile > 67th

asset_regime + market_regime (BTC/USD proxy) both tagged on every trade

ALL REGIME THRESHOLDS ARE CIATS-OWNED: ADX threshold 25, ATR percentile 67th, EMA periods 20/50, rolling window 50 days — all starting values. CIATS validates each after 600 closed trades (100+ per regime bucket).



6. CIATS Architecture and Data Flows

CIATS is fully asynchronous. It never blocks the trading critical path. It reads only from the Logger and writes only to the Parameter Store. Deming PDCA governs every change.

6.1 Two Data Streams

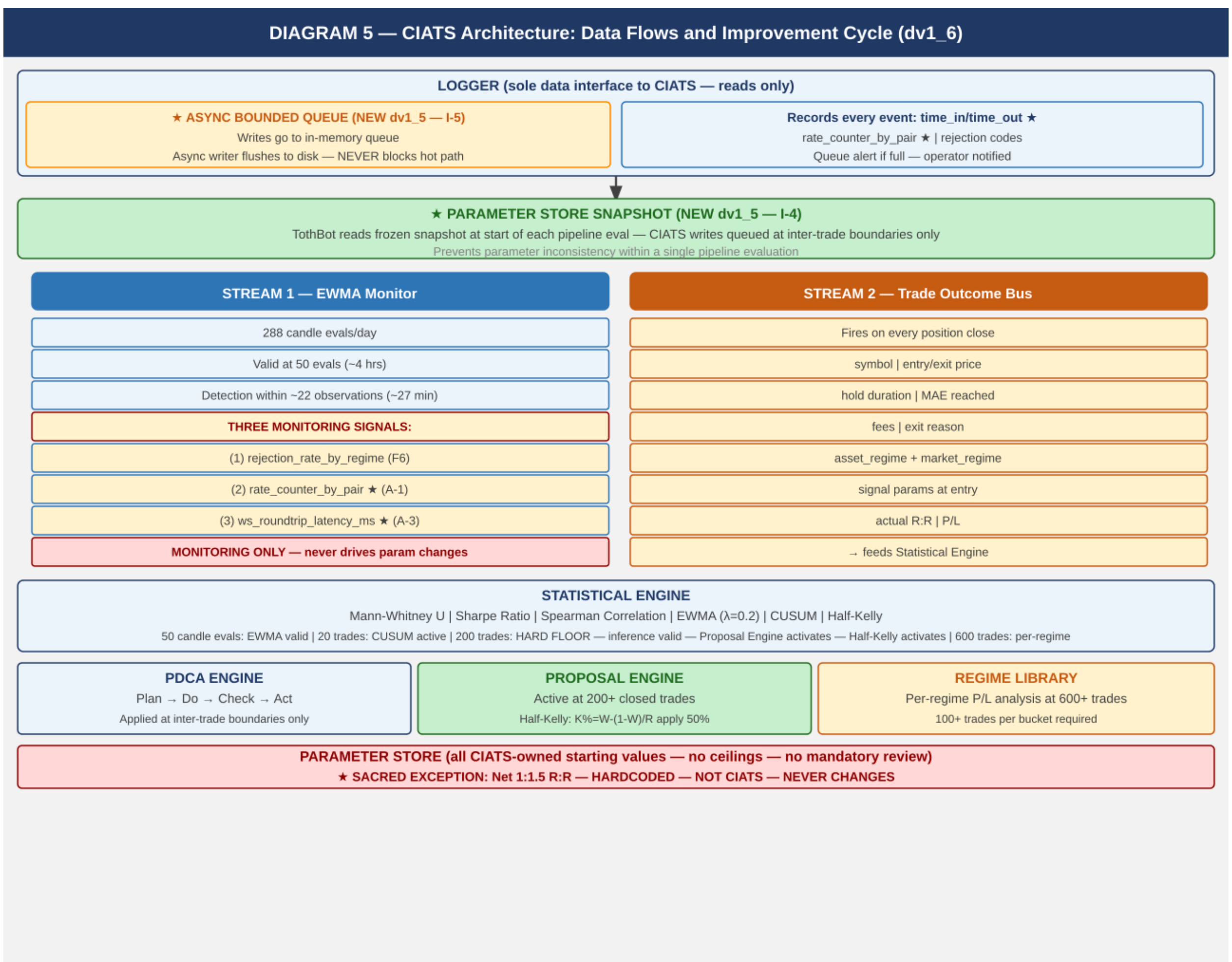
Stream	Description
Stream 1: EWMA Monitor	288 candle evals/day. Valid at 50 evals (~4 hrs). lambda=0.2 [FP/DP validated dv1_6]. Three monitoring signals: (1) post_only rejection_rate_by_regime (F6), (2) rate_counter_by_pair from executions channel (A-1), (3) ws_roundtrip_latency_ms from WS response timestamps (A-3). MONITORING SIGNALS ONLY — never drives parameter changes.
Stream 2: Trade Outcome Bus	Fires on each position close. Captures: symbol, entry/exit price, hold duration, MAE reached, fees, exit reason, both regime tags, signal params at entry, actual R:R, P/L.

LOGGER — ASYNC BOUNDED QUEUE (NEW dv1_5 — I-5): Logger writes go to an in-memory bounded queue. Queue size: 10,000 entries [FP/DP validated dv1_6: 20 positions x 10 events/candle x 50-candle buffer]. A separate async writer flushes to disk. The Logger NEVER blocks the trading hot path under any condition. Queue must be bounded with an immediate alert if queue fills. Assign to 0511006.

PARAMETER STORE — SNAPSHOT PATTERN (NEW dv1_5 — I-4): TothBot takes a frozen snapshot of all CIATS parameters at the start of each pipeline evaluation. The pipeline uses the snapshot throughout. CIATS writes are queued and applied only at confirmed inter-trade boundaries. Prevents parameter inconsistency within a single pipeline evaluation. Assign to 0511002 and 0211001.

6.2 Statistical Thresholds

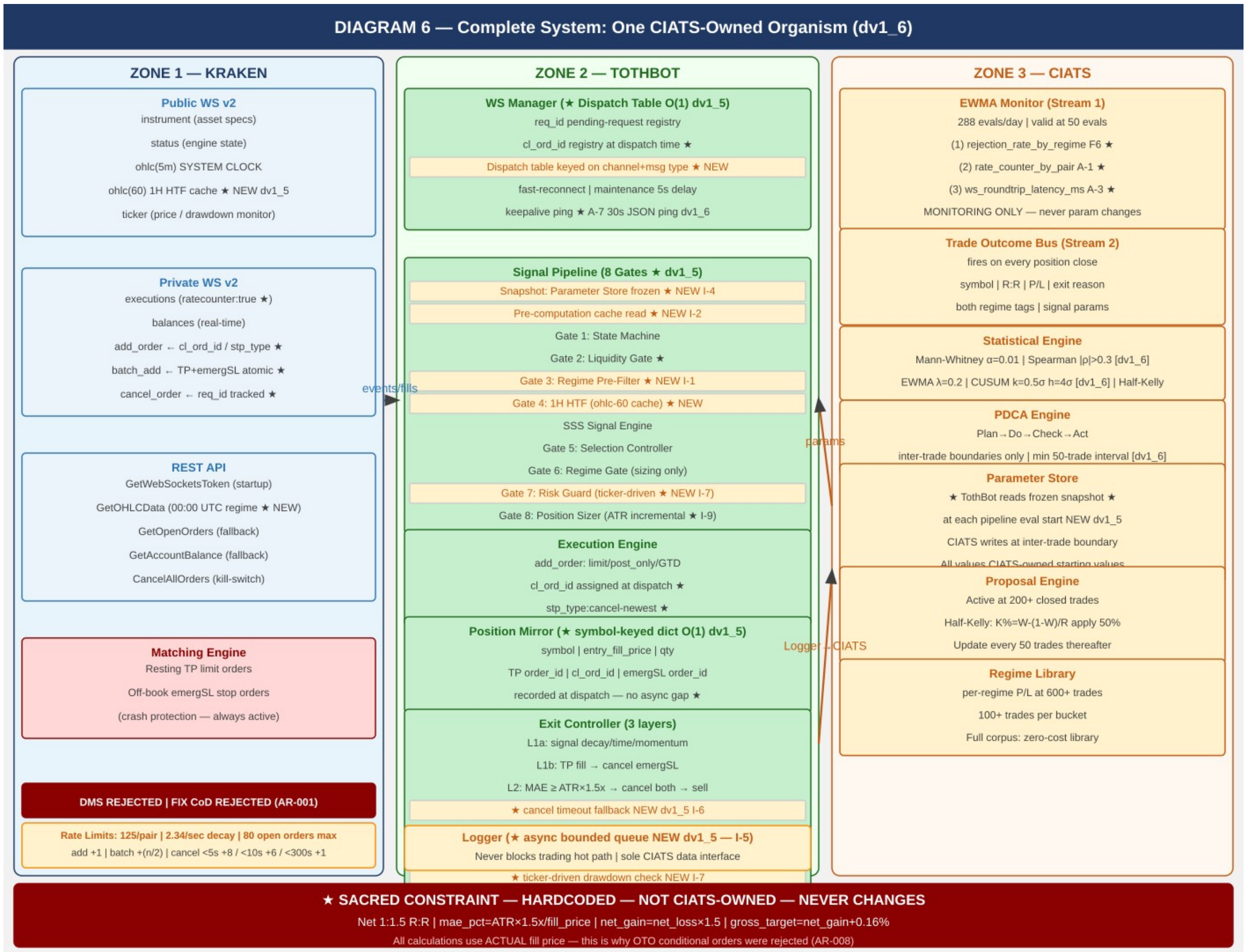
Threshold	Purpose
50 candle evals	EWMA Monitor valid. Fast warning stream active. Monitoring only. lambda=0.2 [validated dv1_6: half-life=3.1 evals].
200 trades — HARD FLOOR	Inference valid. Proposal Engine activates. Half-Kelly replaces fixed 5%. FP/DP validated [dv1_6]: SE(win_rate)~3.5%, SE(Sharpe)~7.2%. Adequate for directional inference.
600 trades	Per-regime analysis: 100+ per bucket. Full regime-segmented optimization. Mann-Whitney power ~85% at alpha=0.01 [FP/DP validated dv1_6].
Rolling 200-300	Current performance window — what the system is doing NOW.
Full corpus	Zero-cost historical library. Grows perpetually. Regime comparison.
Mann-Whitney alpha [dv1_6] ★	alpha=0.01 (one-sided). 99% confidence required for financial parameter decisions. alpha=0.05 produces false positives that increase over-optimization risk.
Spearman rho threshold [dv1_6] ★	|rho| > 0.3 AND p < 0.05 (BOTH conditions required simultaneously). Values below either threshold treated as noise. Use scipy.stats.spearmanr.
CUSUM parameters [dv1_6] ★	Starting values: k=0.5*sigma, h=4*sigma of monitored metric. Subject to CIATS optimization after 200-trade floor. Detects persistent drift in per-trade P&L EWMA.
PDCA minimum interval [dv1_6] ★	Minimum 50 trades between any two CIATS parameter changes after 200-trade floor. Ensures statistical independence between evaluation windows. Prevents over-optimization.



7. Complete System: One CIATS-Owned Organism

This diagram shows all three zones — Kraken, TothBot, and CIATS — as a single unified organism. Every flow and connection point is shown.

CIATS OWNS EVERYTHING EXCEPT 1:1.5 R:R: All parameters, thresholds, sizing decisions, halt thresholds, multipliers, and trading rules are CIATS-owned starting values. They will be replaced by data-driven values during paper trading and live trading.



8. Locked Architectural Decisions

ID	Decision	Specification
AR-001	No Dead Man Switch	Resting TP limit orders and off-book emergency SL stop orders on Kraken matching engine ARE the crash protection. EVALUATED AND REJECTED: (1) cancel_all_orders_after (WS v2 DMS) — would cancel ALL orders on TothBot failure including the protective emergSL. (2) FIX API cancel-on-disconnect — same architectural conflict. Both destroy the crash protection mechanism they are supposed to support.
AR-002	Position Mirror	Symbol-keyed dict for O(1) lookup per ticker event (NEW dv1_5 — I-3). Fields: symbol (key), entry_fill_price, qty, TP order_id, cl_ord_id, emergSL order_id. cl_ord_id recorded at dispatch time before Kraken response — no async gap.
AR-003	Restart reconciliation	PREFERRED: WS executions snap_orders:true on reconnect.

		FALLBACK: REST GetOpenOrders + GetAccountBalance — cold-start only or if snap fails. Verify emergSL orders — re-submit missing — then begin signals.
AR-004	Two API key pairs	DATA key (market data) + TRADE key (orders + WS auth). Nonce window: 10,000ms on both.
AR-005	WS v2 only	WS v1 not used. All future Kraken enhancements go into v2 only.
AR-006	Entry order type	WS add_order: limit, post_only=true, GTD, expiretm=now+30-60s. cl_ord_id assigned by TothBot. stp_type:cancel-newest. CIATS-owned timeout.
AR-007	TP placement	Resting limit via WS batch_add on entry fill. Actual fill price used for R:R calculation. Maker fee 0.16%.
AR-008	Emergency SL placement	Stop-loss market via WS batch_add with TP in single atomic message. ATR(14) x 3.0x from actual fill price. batch_add confirmed correct — OTO conditional evaluated and REJECTED: OTO uses limit price not actual fill price, violates sacred R:R; creates Position Mirror race condition.
AR-009	Short deferred	Kraken Derivatives. Activates when conditions allow.
AR-010	24/7/365	No stand-down. CIATS adds via PDCA only if data supports after 200 trades.
AR-011	Net 1:1.5 R:R — SACRED	Only hardcoded constraint. Dynamic formula using actual fill price. NEVER changed by CIATS.
AR-012	Fee baseline	Maker 0.16% / Taker 0.26%. CIATS monitors actuals.
AR-013	Kraken Pro — Intermediate	Confirmed sufficient for all required API permissions.
AR-014	Logger async bounded queue	Logger writes go to in-memory bounded queue (size: 10,000 entries [FP/DP validated dv1_6]). Separate async writer flushes to disk. Logger NEVER blocks trading hot path. Queue alert if full. Assign to 0511006. (NEW dv1_5 — I-5)
AR-015	WS dispatch table	All incoming WS messages routed via dispatch table keyed on channel+message type. O(1) routing — no if/elif chain. Assign to 0511001. (NEW dv1_5 — I-8)
AR-016	Incremental indicator maintenance	ATR(14) maintained as incremental running calculation updated on each 5m candle close. O(1) per update vs O(N) recomputation. Assign to 0511002, 0511004. (NEW dv1_5 — I-9)
AR-017	Daily regime via REST GetOHLCData	REST GetOHLCData called once per monitored pair at 00:00 UTC. Returns 720 entries. CRITICAL dv1_6: ALWAYS exclude last entry (response[-1]) — current not-yet-committed candle corrupts all indicators. Use only response[:-1]. See 1011012. REST is correct tool for scheduled batch reads. FP/DP verified. Assign to 0511002. (NEW dv1_5 — FLAG 2)
AR-018 (dv1_6) ★	amend_order — Primary Amendment	amend_order is primary. edit_order is DEPRECATED. amend_order 43% cheaper for orders resting <5s (+4 vs +7 rate counter). Preserves queue position unlike cancel+resubmit. Assign to 0411001.
AR-019 (dv1_6) ★	batch_cancel — Multi-Order Cancel	batch_cancel cancels multiple orders in single WS message. MUST use for Full Halt scenario to cancel all open entry orders in minimum messages. Max 50 orders per message. Assign to 0511001, 0511004.
AR-020 (dv1_6) ★	Per-pair Instrument Status Cache	WS Manager maintains symbol-keyed status dict updated from instrument channel. Pairs in reduce_only / work_in_progress / delisted / limit_only are NO-ENTRY before pipeline. Prevents wasted rate counter (+1 per rejected order). Assign to 0511001, 0511002.

9. Complete CIATS Parameter Registry

ALL PARAMETERS ARE CIATS-OWNED STARTING VALUES: CIATS owns all parameters after 200-trade gate. No ceilings. No required safety reviews for sizing changes. Deming PDCA governs every change. 1:1.5 R:R net is the only exception.

Parameter	Starting Value	CIATS Changes When...
tradeable_pct	50% of portfolio	Drawdown analysis shows over/under-exposure.
per_trade_pct	5% -> Half-Kelly at 200 trades	Kelly criterion diverges. Updates every 50 trades thereafter.

max_concurrent	20 positions	Concentration data shows optimal concurrency.
mae_mult (ATR x)	1.5x ATR(14)	Actual MAE distribution clusters above/below. Mann-Whitney alpha=0.01 [dv1_6].
emergency_sl_mult	3.0x ATR(14)	Ratio to MAE must remain >=2x. Tail event distribution.
entry_timeout_sec	30-60s GTD	Skip rate analysis vs fill opportunity. Spearman |rho|>0.3 [dv1_6].
full_halt_drawdown	10% of portfolio	Portfolio volatility distribution analysis.
session_pause_drawdown	5% of portfolio	Session loss distribution. Mann-Whitney alpha=0.01 [dv1_6].
signal_thresholds	SSS gates (derived)	Signal cohort win rates diverge. Mann-Whitney.
adx_threshold	25	Regime classification accuracy vs trade outcomes.
atr_percentile_threshold	67th	Vol regime accuracy vs outcome distribution.
htf_ema_periods	20/50 daily	1H HTF gate rejection vs signal quality outcomes.
min_volume_usd_daily	$500k USD	Post_only fill rate and SL quality vs volume floor.
1:1.5 R:R net	HARDCODED — SACRED	NOT CIATS-OWNED. The only permanent constraint in the system.

10. Six FP/DP Validated System Improvements

Derived from 80+ sources across 3 independent research passes plus two deep validation sessions (20 searches, 144+ sources total) plus 10-search Kraken connection validation [dv1_6] (126 sources). 5+ sources per parameter.

Finding 1: 1H HTF Confirmation Gate [TothBot]

Gap: No layer between daily regime and 5-min entry.

Evidence: 15-20% higher win rates with daily+hourly filter (5 independent sources 2024-2025).

Change: Require 1H EMA(20)>EMA(50) for TRENDING_POSITIVE entries. Cache from ohlc(60) WS subscription (NEW dv1_5). Skip logged as HTF_GATE_REJECTED.

Effect: Removes false signals during 1H pullbacks within daily uptrends. Zero API cost.

Finding 2: ATR Dynamic Stop Distances [TothBot/Exit]

Gap: Fixed 2% stop ignores real-time market volatility.

Evidence: +15% performance, -22% max drawdown vs fixed stops (LuxAlgo backtesting + 4 other sources).

Change: MAE: ATR(14) x 1.5x. Emergency SL: ATR(14) x 3.0x. Both CIATS-owned starting values. ATR maintained as incremental running calculation (NEW dv1_5 — I-9).

Effect: Stop adapts to conditions. R:R formula preserved. Exit Controller fires before emergency SL.

Finding 3: Regime Gate [TothBot]

Gap: Long momentum system trading in absent-edge conditions.

Evidence: Accuracy 65%->45-50% in NON_DIRECTIONAL (Market Cipher). Break-even 40% for 1:1.5 R:R.

Change: NON_DIR+ELEV and TRENDING_NEG: no entry (Gate 3 pre-filter — NEW dv1_5). NON_DIR+NORM: 50% size (Gate 6). Daily candles via REST GetOHLCData 00:00 UTC — CRITICAL: exclude response[-1] [dv1_6].

Effect: Stops trading when documented edge is absent. Pure loss minimization.

Finding 4: Minimum Liquidity Gate [TothBot]

Gap: No minimum 24h volume on asset selection.

Evidence: $1M+/day minimum for clean execution (Coin Bureau + 4 sources 2024-2025).

Change: Asset Registry adds min_24h_volume_usd $500k starting. Checked daily via ticker.

Effect: Protects fee formula. Ensures clean post_only fills and SL execution.

Finding 5: CIATS Half-Kelly Pathway [CIATS]

Gap: Fixed 5% sizing had no defined data-driven transition.

Evidence: Half-Kelly: 75% optimal growth at 50% less drawdown (MacLean-Thorp-Ziemba + 4 sources). FP/DP validated [dv1_6].

Change: At 200 trades: K%=W-(1-W)/R, apply 50%. Update every 50 trades. No ceilings.

Effect: Data-driven sizing replaces arbitrary fixed fraction after sufficient data.

Finding 6: post_only Rejection Monitoring [CIATS]

Gap: Rejection events logged but unused by CIATS EWMA.

Evidence: Rejection rate = direct signal of entry placement quality (Cube Exchange + 4 sources).

Change: CIATS EWMA adds rejection_rate_by_regime stream. High rate in TRENDING = miscalibrated.

Effect: Free diagnostic signal already in Logger. Entry optimization pathway activated.

11. Complete Trade Lifecycle — Narrative Reference

11.1 Normal Trade (Entry to TP or Exit Controller)

OHLC 5-min candle close fires on public WS v2

Parameter Store snapshot frozen — all CIATS params captured for this eval (NEW dv1_5 — I-4)

Pre-computation cache read — regime, 1H EMA, ATR(14) incremental, 24h liquidity (NEW dv1_5 — I-2)

★ Per-pair instrument status check — skip pairs in reduce_only/work_in_progress/delisted/limit_only [dv1_6]

Gate 1: State Machine — online or post_only: continue; else halt and wait

Gate 2: Liquidity Gate — 24h volume >= $500k: continue; else skip

Gate 3: Regime Pre-Filter — TRENDING_NEG or NON_DIR+ELEV: STOP; else continue (NEW dv1_5 — I-1)

Gate 4: 1H HTF Gate — 1H EMA(20) > EMA(50) for TRENDING_POS: continue; else skip (cache from ohlc-60)

Signal Engine: SSS score computed on 5-min candle

Gate 5: Selection Controller — 6-gate quality evaluation: pass or reject

Gate 6: Regime Gate — TRENDING_POS: proceed; NON_DIR+NORM: 50% size

Gate 7: Risk Guard — drawdown/concentration/exposure checks: pass or halt

Gate 8: Position Sizer — MAE = ATR x 1.5x (incremental), emergSL = ATR x 3.0x, TP = R:R formula

Execution: WS add_order (limit, post_only, GTD expiretm=now+30-60s, cl_ord_id assigned, stp_type:cancel-newest)

If expired/rejected: log reason code -> CIATS EWMA monitors rejection rate. Done.

If filled: WS batch_add -> TP (resting limit) + emergSL (stop-market) atomic. cl_ord_id recorded.

Position Mirror records: symbol (key), entry_fill_price, qty, TP order_id, cl_ord_id, emergSL order_id

Exit Controller monitors via ticker (continuous drawdown check — NEW dv1_5 — I-7)

Logger records full entry via async queue (time_in/time_out). CIATS Trade Outcome Bus ready.

TP fills: EC cancels emergSL. Or EC fires Layer 1/2 exit. Position closes.

Logger records full exit. CIATS Trade Outcome Bus fires with complete trade record.

11.2 Layer 2 MAE Exit (Adverse Price)

Exit Controller detects: current_price - entry >= ATR(14) x 1.5x (from incremental ATR cache)

WS cancel_order: cancel TP resting order

WS cancel_order: cancel emergSL stop order (off-book) -- SEQUENCE CRITICAL: both cancels before market sell

CANCEL TIMEOUT FALLBACK (NEW dv1_5 — I-6): req_id registry tracks each cancel ACK. On timeout: check executions channel state. If order confirmed cancelled: proceed to next step. If state unknown: retry once. If second timeout: ALERT and hold position — alert operator immediately. NEVER issue market sell with ambiguous order state. Assign to 0511001 (WS Manager timeout behavior) and 0511004 (Exit Controller fallback logic).

WS add_order: market sell (taker 0.26%)

MPP NOTE (C-1 dv1_4): Market sell subject to MPP enforcement. May reject if bid/ask spread wide during adverse move. Exit Controller MUST handle MPP rejection and retry. Specification assigned to 0511004.

Logger: exit reason = MAE_THRESHOLD_BREACH

CIATS Trade Outcome Bus fires

11.3 Emergency SL (Layer 3 — System Failure Only)

VPS crashes / TothBot dies / internet severs / unrecoverable WS disconnect

Kraken matching engine fires off-book emergency SL stop-market order at ATR x 3.0x

MPP NOTE (C-1 dv1_4): The triggered market order is subject to MPP enforcement. May reject on wide spread at time of trigger. Specification assigned to 0411001.

On restart: reconciliation detects closed position -> Logger: EMERGENCY_SL_FIRED

This is a system failure event requiring investigation. SHOULD NEVER FIRE IN NORMAL OPERATION.

12. Open Items — Require Specification Documents

#	Item	Description
001	Signal generation (SSS)	SSS scoring model re-derived from first principles for Kraken Spot.
002	Asset selection universe	Kraken USDT/USDC pairs — eligibility criteria beyond liquidity gate.
003	Exit Controller spec	Complete: signal decay, time exit, momentum detection, full L1/L2 decision tree. MPP rejection handling (C-1). Cancel timeout fallback (I-6). batch_cancel for Full Halt (GAP-05 dv1_6). OTO evaluated and rejected — batch_add confirmed correct.
004	Entry limit placement	Exact rule for post_only limit price relative to bid/ask. Evaluate ticker event_trigger:bbo. Evaluate L2 book (PL-010). Evaluate L3 wss://ws-l3.kraken.com/v2 (PL-009 — URL corrected dv1_6).
005	Selection Controller	6-gate formal spec. Must include per-pair status check (GAP-07 dv1_6).
006	CIATS specification	Full spec: all interfaces, methods, PDCA workflows, regime library. Parameter Store snapshot. Async Logger. Mann-Whitney alpha=0.01, Spearman |rho|>0.3, CUSUM k/h, PDCA 50-trade interval [dv1_6].
007	AA=04 Exchange Interface	Formal controlled document incorporating all decisions in this overview.
00	All AA=10 coding specs	Full coding specs for every module. WD documents created this session: 1011001, 1011002, 1011010, 1011012 [dv1_6].

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13. Revision History

Version	Date	Description
dv1_0	03-23-2026	Initial document. All architectural decisions, FP/DP improvements, exit architecture ground zero redesign, regime taxonomy, CIATS ownership model, and complete system flow.
dv1_1	03-23-2026	Added all six system flow diagrams.
dv1_2	03-24-2026	Six changes from deep API validation search (TB00003): C-1 emergency SL terminology, C-2 rate limits, C-3 OHLC stall, A-1 ratecounter, A-2 req_id, A-3 latency logging.
dv1_3	03-24-2026	Rebuilt as correct .docx with all six diagrams embedded. dv1_2 produced in error without diagrams.
dv1_4	03-24-2026	Seven changes from second deep API validation search (TB00005): C-1 MPP, A-1 AR-001 expansion, A-2 cl_ord_id, A-3 snap_orders, A-4 stp_type, A-5 reconnect distinction, A-6 OTO rejection.
dv1_5	03-24-2026	Eleven changes: FLAG 1 ohlc(60) for 1H HTF, FLAG 2 REST GetOHLCData daily regime, I-1 Gate 3 Regime Pre-Filter, I-2 pre-computation cache, I-3 Position Mirror O(1) dict, I-4 Parameter Store snapshot, I-5 Logger async bounded queue, I-6 cancel timeout fallback, I-7 ticker-driven drawdown, I-8 WS dispatch table, I-9 ATR incremental. All six flow diagrams updated.
dv1_6	03-24-2026	Six changes from 10-search FP/DP Kraken connection validation + FP/DP Operations/CIATS analysis (TB00007). (1) amend_order primary, edit_order deprecated; batch_cancel added for Full Halt; L3 URL corrected to wss://ws-l3.kraken.com/v2. (2) Rate table: amend_order row added with reduced cost. (3) A-7 WS ping strategy (30s JSON ping, 10s timeout); A-8 zombie detection (90s threshold); A-9 balances sequence gap detection; snap_orders/snap_trades corrected (snapshot deprecated). (4) Per-pair instrument status check added before Gate 1 — instrument channel cache. AR-018, AR-019, AR-020 added. (5) GetOHLCData response[-1] exclusion — ALWAYS not-yet-committed candle — CRITICAL correctness requirement. (6) Auth verification: Oct 1 2025 breaking change — verify before coding. CIATS statistical: Mann-Whitney alpha=0.01, Spearman |rho|>0.3 p<0.05, CUSUM k=0.5s h=4s, PDCA 50-trade interval, Logger queue 10,000 entries — all FP/DP validated. All six diagrams updated.